
Multimedia Appendix

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Multimedia Appendix 2

Three-Pillar Organizational Assessment Rubric

This appendix presents the full 3-pillar assessment rubric proposed in the main paper (section “The 3-Pillar Rubric”). Each indicator is scored as low, medium, or high strength using evidence-based anchors from the literature reviewed in the main paper’s section “The Evidence Base: 3 Pillars”. Organizations scoring predominantly “low strength” across multiple pillars face the self-reinforcing degradation cycle that the framework identifies as the central threat.

Pillar 1: Analytics Maturity

Indicator	Low strength	Medium strength	High strength	Evidence
HIMSS AMAM stage	Stages 0-2: Fragmented data, limited reporting	Stages 3-4: Integrated warehouse, standardized definitions	Stages 5-7: Predictive analytics, AI integration	[1,2]
Self-service analytics	None; all analytics require IT intervention	Partial; BI tools available but underutilized	Widespread; clinical staff access data directly	[3,4]
AI and NL interface	No NL2SQL or conversational analytics	Pilot programs or evaluation underway	Natural language query capability deployed	[5,6]

Pillar 2: Workforce Agility

Indicator	Low strength	Medium strength	High strength	Evidence
First-year analytics staff turnover	>30% (high instability)	15-30%	<15% (high stability)	[7,8]
Average leadership tenure	< 3 years	3-5 years	> 5 years	[9]
Knowledge concentration	Critical expertise held by 3 or fewer individuals	Partial documentation; some cross-training	Distributed expertise; documented processes	[10,11]

Pillar 3: Technical Enablement

Indicator	Low strength	Medium strength	High strength	Evidence
Data access	SQL/technical expertise required for all queries	IT queue for complex queries; basic self-service	Natural language or visual query interfaces	[4,6]
Interoperability	Fragmented systems; manual reconciliation	Partial integration; some automated feeds	Unified data platform; real-time integration	[12,13]
Schema coupling	Hard-coded reports break on schema change	Partial semantic layer; some automated feeds	Semantic layer adapts; CI/CD detects drift	[14,15]

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